

One model, one complete corpus epoch

Train contacts-v1 Qwen3 1.5B from scratch on all native and ProteinMPNN-redesigned AFDB and ESM-Atlas documents. Companion to Zack's broader issue #274 sweep; tracked in issue #277.

Full-corpus coverage

232,090,905 documents and 248.584B raw tokens produce 34,092,146 packed examples. Concatenate the four complete caches, shuffle, and visit each example exactly once. No native/redesigned mixture weighting. One epoch is 266,345 updates at batch 128; the final batch has 114 real and 14 padding examples.

Optimizer and runtime

Retain exp232 m2-p06: LR 0.001, weight decay 0.2, sequence length 8192, blocked attention, and scheduled amino-acid order augmentation. WSD warmup 10%, decay 20%, minimum LR ratio 0.1 spans the full epoch. Estimated duration: 66–68 hours on 128 H100s at batch priority in cw-us-east-02a.

Earlier run retained

The original 50:50 weighted-mixture run was stopped intentionally at about step 69,556 on 2026-09-10 at 13:38 UTC when the requested objective changed. Latest validation loss: 3.10302. Permanent checkpoints through step 58,080 remain available. No production failures. Its metrics and history remain a separate record.

Validation and current status

All four full caches are verified. The in-region packing audit used the trainer's exact implementation. Four tests passed, including finite coverage across unequal source sizes and a partial shuffle block. The existing document-length policy trims one trailing token in 862 documents; every document remains represented. The full-corpus ten-step GPU smoke succeeded with finite losses and native/HF checkpoints including the tokenizer. Production reached step 73,702 / 266,345 (27.67%) on September 11. All 34 validation passes improved, reaching 3.08621 at step 71,876. The first failure occurred at 09:10 UTC during a checkpoint save: distributed shutdown timed out after worker 6 began shutdown; initiating cause unconfirmed. Step 72,744 is the last completed recovery checkpoint. Attempt a03 launched at 09:17:36 UTC; all 16 workers are running at batch priority. Checkpoint restore succeeded and training resumed from step 72,745 at 09:27 UTC with finite loss around 2.88. Post-restart checkpoint 73,199 saved successfully. W&B caught up by 09:46 UTC: step 73,958 (27.77%), loss 2.87417, 0.877 seconds/update. Same W&B run: contacts-v1-exp277-m2-p06-full-epoch-1.5B. Training remains in the stable LR phase. Permanent checkpoints 26,634 and 53,268 are preserved. Continuous monitoring is active.

Interpretation

One seed provides an initial answer to the redesign question. Compare with the native-only winner while reporting the different training exposure. Contact-prediction quality has not yet been evaluated.